Functional data analysis

8. Functional time series

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Objective are:

- *to introduce some functional time series models, including functional linear processes;*
- *to present some estimation methods and hypothesis testings;*
- *to present a case study.*

Outline of the lecture

1. *Introduction*
2. *Functional autoregressive process*
3. *Case study*

Introduction

- We shall consider functional data observed sequentially over time.
- For example, the curves $X_1, X_2, \dots,$ can represent pollution levels at some location;
- $X_n(t)$ is the level of pollution at time t of day n.
- Such curves cannot be treated as being independent with the same distribution; they do not form a simple random sample of curves.
- Pollution on a given day will be affected by pollution on previous days, so the curves will be dependent.
- A daily pollution pattern will be different on Sunday and on Monday, so the curves will not have the same distribution every day.

Functional autoregressive process

Definition

We say that a sequence $\{X_n,\ -\infty < n < +\infty\}$ of mean zero elements of L_2 follows a functional (mean zero) AR(1), FAR(1), model if

$$X_n = K(X_{n-1}) + \varepsilon_n,$$

where

- $K : L_2 \to L_2$ is a linear operator transforming a function into another function, and
- $\{\varepsilon_n, -\infty < n < \infty\}$ is a sequence of iid mean zero errors.

- In applications, the operator K is an integral operator defined by

$$K(x)(t) = \int_0^1 \kappa(t,s)x(s)ds, \quad x \in L_2.$$

- Equations can thus be rewritten more directly as

$$X_n(t) = \int_0^1 \kappa(t,s)X_{n-1}(s)ds + \varepsilon_n(t).$$

- To simplify the formulas, we also assume that the mean function of each X_n is zero.
- The general FAR(1) process is defined by the equation

$$X_n - \mu = K(X_{n-1} - \mu) + \varepsilon_n.$$

- If μ is unknown one uses its estimator $\widehat{\mu}$.

Definition

We say that a sequence $\{X_n,\ -\infty < n < +\infty\}$ of L_2 random functions is (weakly) stationary if

- $E(X_n) = 0$ for any n;
- $E(X_{n+h} \otimes X_{m+h}) = E(X_n \otimes X_m)$ for any n, m, h.

Definition

For a stationary sequence $\{X_n,\ -\infty < n < +\infty\}$ of L_2 random functions we define lag-h autocovariance operator $Q_h : L_2 \to L_2$ by

$$Q_h = E(X_n \otimes X_{n+h})$$

which acts so that

$$\langle Q_h x, y \rangle = E(\langle X_n, x \rangle \langle X_{n+h}, y \rangle), \quad x, y \in L_2.$$

Existence of stationary FAR(1) sequence

Theorem

If $K : L_2 \to L_2$ is linear bounded operator and

$$||K|| := \sup_{||x|| \leq 1} ||Kx|| < 1,$$

then there is a unique stationary causal solution to

$$X_n = KX_{n-1} + \varepsilon_n$$

given by

$$X_n = \sum_{j=0}^{\infty} K^j \varepsilon_{n-j}.$$

The series converges almost surely, and in the L_2 sense, i.e.

$$E\left\| X_n - \sum_{j=0}^{m} K^j \varepsilon_{n-j} \right\|^2 \to 0 \text{ as } m \to \infty.$$

Example

Consider an integral operator on $L_2(0,1)$ defined by

$$Kx(t) = \int_0^1 \kappa(t,s)x(s)ds, \quad x \in L_2(0,1)$$

which satisfies

$$\int_0^1 \int_0^1 \kappa^2(t,s)dtds < 1.$$

- For this example we have

$$||K|| < 1.$$

Esimation

Let (X_n) be a stationary sequence of L_2 random elements that follows FAR(1) model

$$X_n = KX_{n-1} + \varepsilon_n.$$

- Multiplying by X_{n-1} we get

$$X_n \otimes X_{n-1} = KX_{n-1} \otimes X_{n-1} + \varepsilon_n \otimes X_{n-1}$$

- and taking expectations yield

$$E(X_n \otimes X_{n-1}) = E(KX_{n-1} \otimes X_{n-1}) + E(\varepsilon_n \otimes X_{n-1}).$$

- Defining the lag-1 autocovariance operator by

$$Q_1 = E(X_n \otimes X_{n-1}), \quad Q_1 x = E(\langle X_n, x\rangle X_{n-1}), \quad x \in L_2,$$

  we have the identity

$$Q_1 = KQ.$$

- The above identity is analogous to the scalar case, so we would like to obtain an estimate of K by using a finite sample version of the relation

$$K = Q_1 Q^{-1}.$$

- The operator Q does not however have a bounded inverse on the whole of L_2.
- To see it, recall that Q admits representation

$$C(x) = \sum_{j=1}^{\infty} \lambda_j \langle x, v_j \rangle v_j, \quad x \in L_2,$$

which implies that $Q^{-1} Q x = x$, where

$$Q^{-1} x = \sum_{j=1}^{\infty} \lambda_j^{-1} \langle x, v_j \rangle v_j.$$

- The operator Q^{-1} is defined if all λ_j are positive.

- A practical solution is to use only the first p most important EFPC's $\widehat{v}_j$, and to define

$$\widehat{K}_p x = \sum_{j=1}^{p} \widehat{\lambda}_j^{-1} \langle x, \widehat{v}_j \rangle \widehat{v}_j.$$

- The operator $\widehat{K}_p$ is defined on the whole of L_2, and it is bounded if $\lambda_j > 0$ for $j = 1, \ldots, p$.
- To derive a computable estimator of K, we use an estimator of Q_1 defined by

$$\widehat{Q}_1 x = \frac{1}{n-1} \sum_{k=1}^{n-1} \langle X_k, x \rangle X_{k+1};.$$

- we obtain, for any $x \in L_2$,

$$\begin{aligned}\widehat{Q}_1\widehat{K}_p &= \widehat{Q}_1\Big(\sum_{j=1}^{p} wh\lambda_j^{-1}\langle x, \widehat{v}_j\rangle\widehat{v}_j \\ &= \frac{1}{n-1}\sum_{k=1}^{n-1}\langle X_k, \sum_{j=1}^{p}\widehat{\lambda}_j^{-1}\langle x, \widehat{v}_j\rangle\widehat{v}_j\rangle X_{k+1} \\ &= \frac{1}{n-1}\sum_{k=1}^{n-1}\sum_{j=1}^{p}\widehat{\lambda}_j^{-1}\langle x, \widehat{v}_j\rangle\langle X_k, \widehat{v}_j\rangle X_{k+1}.\end{aligned}$$

- The estimator $\widehat{Q}_1\widehat{K}_p$ can be used in principle, but typically an additional smoothing step is introduced by using the approximation $X_{k+1} \approx \sum_{i=1}^{p}\langle X_{k+1}, \widehat{v}_i\rangle\widehat{v}_i$.

- This leads to the estimator

$$\widehat{K}_p x = \frac{1}{n-1} \sum_{k=1}^{n-1} \sum_{j=1}^{p} \sum_{i=1}^{p} \widehat{\lambda}_j^{-1} \langle x, \widehat{v}_j \rangle \langle X_k, \widehat{v}_j \rangle \langle X_{k+1}, \widehat{v}_i \rangle \widehat{v}_i.$$

- This estimator is a kernel operator with the kernel

$$\widehat{\kappa}_p(t,s) = \frac{1}{n-1} \sum_{k=1}^{n-1} \sum_{j=1}^{p} \sum_{i=1}^{p} \widehat{\lambda}_j^{-1} \langle X_k, \widehat{v}_j \rangle \langle X_{k+1}, \widehat{v}_i \rangle \widehat{v}_j(s) \widehat{v}_i(t).$$

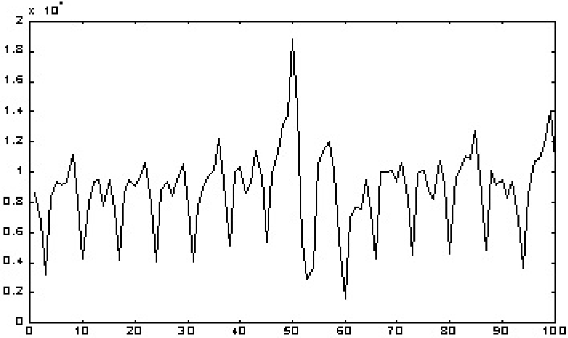
- This is verified by noting that

$$\widehat{K}_p x(t) = \int_0^1 \widehat{\kappa}_p(t,s) x(s) ds.$$

- All quantities entering the setimator $\widehat{\kappa}_p$ are available as output of the `R` function `pca.fd`, so this estimator is very easy to compute.

Case study

- The problem of mathematical prediction of transactions dynamics in ATM and POS networks is considered.
- Data constitute transactions made in ATM and POS networks between 03/11/2000 and 10/02/2001 and were extracted from the data warehouse.



Number of transactions over 100 days in ATM network.

- One day is normalized to be the interval $[0, 1]$.
- The number of transactions is interpreted as double stochastic Poisson process $(N^{(q)}(t), t \in [0, 1])$, where $q = 1$ corresponds to ATM network whereas $q = 2$ corresponds to POS network.
- The corresponding conditional intensities $\Lambda^{(q)}(t), t \in [0, 1]$ given by

$$\Lambda^{(q)}(t) = \int_0^t \lambda^{(q)}(s)ds = E[N^{(q)}(t)|\mathcal{F}]$$

  are stochastic processes for which we have observations

$$\{\Lambda_i^{(q)}(t_j),\ i = 1, \ldots, n, j = 0, 1, \ldots, 1024\}, \quad q \in \{1, 2\}.$$

- Here n denotes the number of days which are analyzed, and it equals to either 100 or 86 after censoring (removing two weeks of Christmas effect). In what follows $t_j = j/1024, j = 0, 1, \ldots, 1024$.

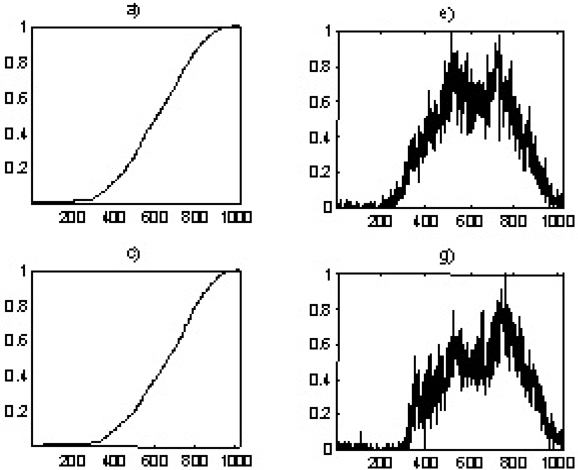
- One day (more precisely twenty-four hours) is divided into 1024 equal time intervals.
- Motivation for choosing this number is that first, an empirical process is constructed by counting the number of transactions during a time interval approximately equal to one minute, and second, the number of intervals is reasonable to be equal to a power of two.
- Using the observations of the intensity functions $\Lambda^{(q)}(t)$, we build the observations for its derivative $\lambda^{(q)}(t)$. Namely, we consider

  $$\{\lambda_i^{(q)}(t_j),\ i = 1, \ldots, n; j = 1, \ldots, 1024\},$$

  where

  $$\lambda^{(q)}(t_j) = \Lambda^{(q)}(t_j) - \Lambda^{(q)}(t_{j-1}), \quad j = 1, \ldots, 1024.$$

- Figure below displays the data of one randomly chosen day.



The numbers of transactions and the intensities of transactions.

Choice of a functional subspace

In what follows $(x_i(t_j))$ will denote either of the four sets of observations: $\Lambda_i^{(q)}(t_j)$, $q = 1, 2$ $\lambda_i^{(q)}(t_j)$, $q = 1, 2$.

- Next step is to convert a raw functional data $(x_i(t_j))$ into a true functional form $(y_i(t), t \in [0, 1])$.
- To this aim smoothing by function bases is used. Hence,

$$y_i(t) = \sum_{j=1}^{m} c_{ij} B_j(t),$$

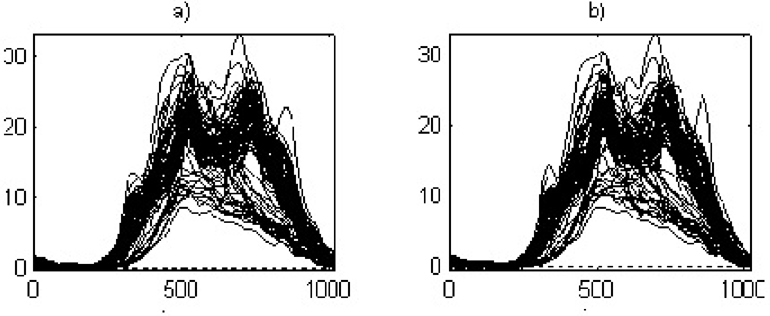
  where $(B_j(t))$ are either cubic B-splines or Fourier bases.
- By choosing the bases we are making hypothesis that our functions of interest can be approximated either by polynomials (in the case of B-splines) or by a fixed number of frequencies (in the case of Fourier bases).

- One of the key issues at this step is the number of coefficients m one has to choose.
- one motivation for $m = 32$ coefficients is that firstly $1/32$ of a day represents 45 minutes and secondly $32 = \sqrt{1024}$. One can check that in many cases square root of the number of observations for the dimension gives optimal, in a sense, approximation.

- Coefficients $C = (c_{ij})$ were found by the least square method:

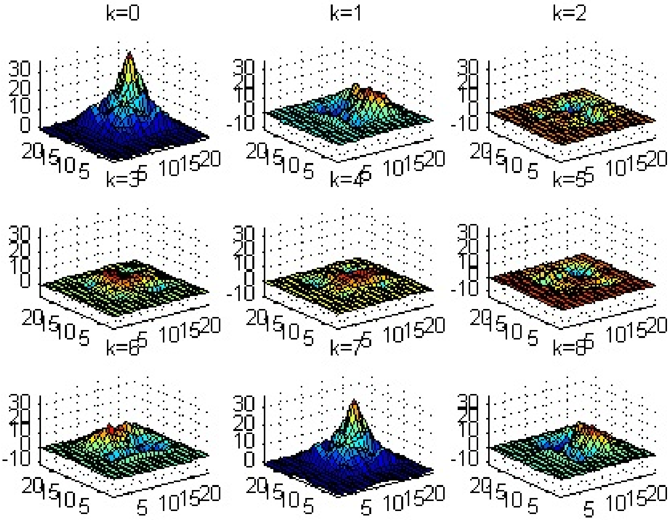
$$C = (B'B)^{-1}B'X,$$

  where $B = (B_j(t_l), j = 1, \ldots, 32; l = 1, \ldots, 1024)$ is the bases matrix and $X = (x_i(t_l), i = 1, \ldots, 32; l = 1, \ldots, 1024)$ is the data matrix.
- Figure below displays functional observations of ATM network: a) intensity observations smoothed with B-splines b) intensity observations smoothted with Fourier bases.



- As we can see, there is no difference on the first glance between our data presented in B-splines or Fourier bases.
- Similar situation occurs for POS network.
- So in the rest we will consider functional data $(y_i(t), t \in [0,1];\ i = 1, \dots, n)$ obtained smoothing by B-splines only.

Covariance and cross-covariance analysis

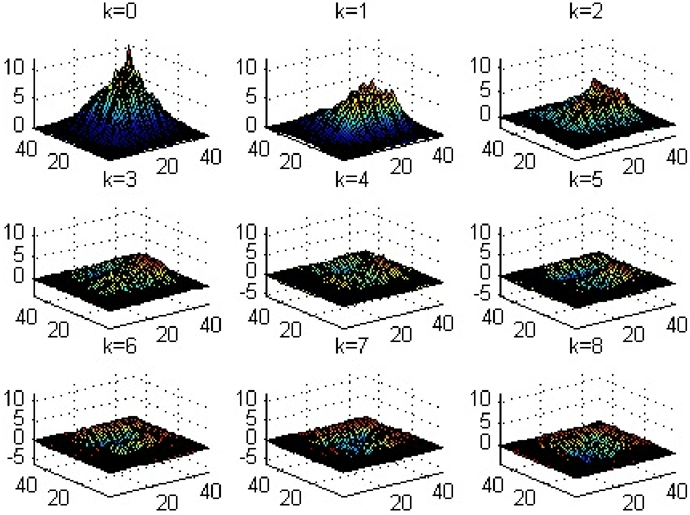


Kernel $q_k(s,t)$ of lag-k autocovariance operator.

- Covariance and cross-covariance analysis of the functional time series $(y_i(t), t \in [0,1], i = 1, \dots, n)$ suggest for differentiation of the functional time series at lag 7.
- Differentiating point-wise we obtain

$$z_i(t) = y_i(t) - y_{i-7}(t), t \in [0,1], \quad i = 1, \dots, n.$$

- Now we consider functional time series $(z_i(t))$ as observations of a stationary sequence in the space $L_2(0,1)$.



Kernel $q_k(s, t)$ of lag-k autocovariance operator of new time series.

Principal component analysis

- The estimate of the covariance function

$$\widehat{h}_z(s,t) = \frac{1}{n}\sum_{k=1}^{n} z_k(s)z_k(t), \quad s,t \in [0,1]$$

defines the operator $\widehat{h}_z : L_2(0,1) \to L_2(0,1)$ by

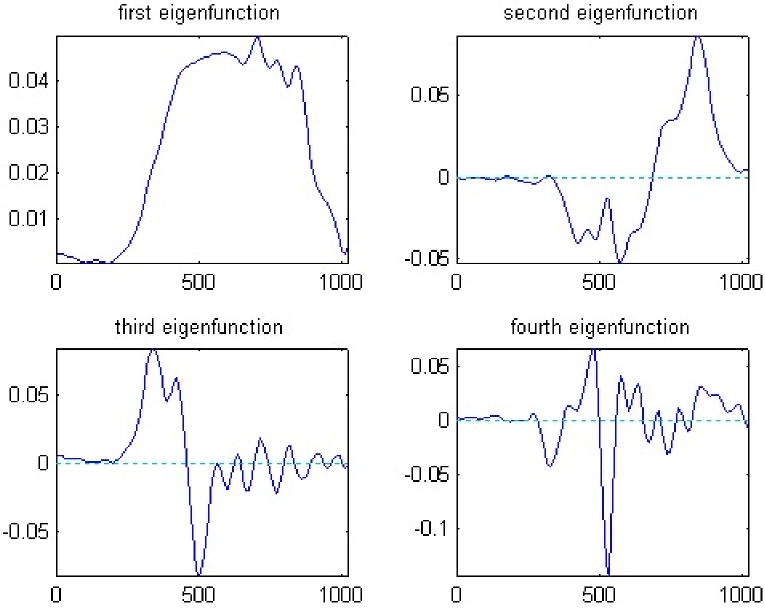
$$\widehat{h}_z u(t) = \int_0^1 \widehat{h}_z(s,t)u(s)ds, \quad t \in [0,1].$$

- This operator is self-adjoint, nonnegative and admits the spectral decomposition

$$\widehat{h}_z = \sum_j \lambda_j \phi_j \otimes \phi_j, \tag{1}$$

where (ϕ_j) is the orthogonal basis in the space $L_2(0,1)$ consisting of eigenfunctions of the operator $\widehat{h}_z$ and (λ_j) denotes the corresponding eigenvalues assuming that $\lambda_1 > \lambda_2 > \cdots$

The first four eigenfunctions are shown in figure:



- When eigenfunctions and corresponding eigenvalues are found one can represent observations in the new bases, and thus obtaining

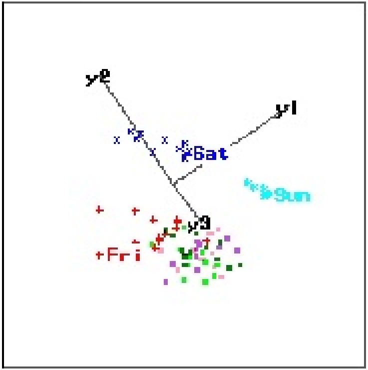
$$\widehat{z}_k(t) = \sum_{j=1}^{M} \lambda_{jk}\phi_j(t), \quad j = 1, \dots, n,$$

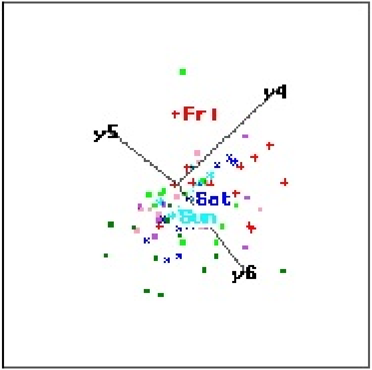
where

$$\lambda_{jk} = \int_0^1 z_k(t)\phi_j(t)dt, \quad k = 1, \dots, n; j = 1, \dots, M.$$

- The crucial point here is the choice of the dimension M.
- Several arguments can be used for this task.

- In the left side of the figure below one see the functional time series $(z_i(t))$ projected onto the first three eigenfunctions of the covariance operator whereas in the right side of the figure, projected onto the next three eigenfunctions.





- Another technique that can be used to estimate a dimension is based on the white noise test of functional data representation in each eigenfunction successively.
- One starts with $M = 1$ and tests the following null hypothesis: the time series $\{\lambda_{11}, \lambda_{12}, \ldots, \lambda_{1n}\}$ follows a white noise.
- Next one tests the null hypothesis for the time series $\{\lambda_{21}, \lambda_{22}, \ldots, \lambda_{2n}\}$ ect.
- The procedure stops when coefficients starts to behave like a white noise.
- All used statistics (including Kolmogorov-Smirnov statistic) rejected the null hypothesis for the first four eigenfunctions.
- Finaly, we used the *ex poste* prediction to find the best in a sense dimension for the functional data $(\widehat{z}_k(t))$ (see the next section below).
- In all approaches the dimension $M = 4$ was found to be satisfactory, and has been used for the final model.

Estimating FAR(1) model

- Consider for the functional time series $(z_i(t), t \in [0,1]; i = 1, \dots, n)$ the functional auto-regressive model of the first order (FAR(1)): for $i = 1, \dots, n$

$$z_i(t) = \int_0^1 \beta(s,t) z_{i-1}(s) ds + \varepsilon_i(t) \quad \text{for each} \quad t \in [0,1],$$

  where it is assumed that

$$\int_0^1 \int_0^1 \beta^2(s,t) ds dt < \infty.$$

- The error terms $(\varepsilon_i(t), t \in [0,1])$ are assumed to be independent random functions such that $E\varepsilon_i(t) = 0$ for each $t \in [0,1]$; $E \int_0^1 \varepsilon_i^2(t) dt = \sigma^2 < \infty$ and $E \int_0^1 \varepsilon_i(t) z_j(t) dt = 0$ for $j < i$.

- To estimate the function $\beta(s,t), s,t \in [0,1]$ we proceed as follows.
- Since (ϕ_j) the eigenfunctions of the covariance operator constitute the orthonormal bases in the spase $L_2(0,1)$ we can represent

$$\beta(s,t) = \sum_{j,k=1}^{\infty} d_{jk}\phi_j(s)\phi_k(t), \quad s,t \in [0,1].$$

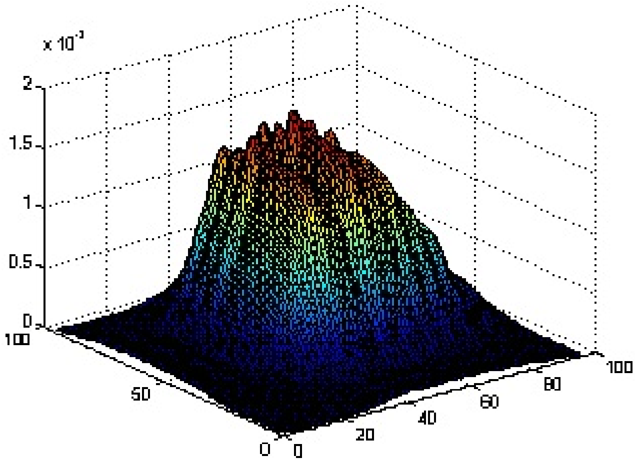
- Thus, our estimator of the function β is given by

$$\widehat{\beta}(s,t) = \sum_{u,v=1}^{M} \widehat{d}_{uv}\phi_u(s)\phi_v(t), \quad s,t \in [0,1],$$

where

$$\widehat{d}_{uv} = \frac{(n-1)^{-1}\sum_{i=1}^{n-1}\langle z_{i-1},\phi_v\rangle\langle z_i,\phi_u\rangle}{n^{-1}\sum_{i=1}^{n}\langle z_i,\phi_v\rangle\langle Z_i,\phi_u\rangle}$$

is a consistent estimator of the coefficient d_{uv}. Figure below shows the function $\widehat{\beta}$.



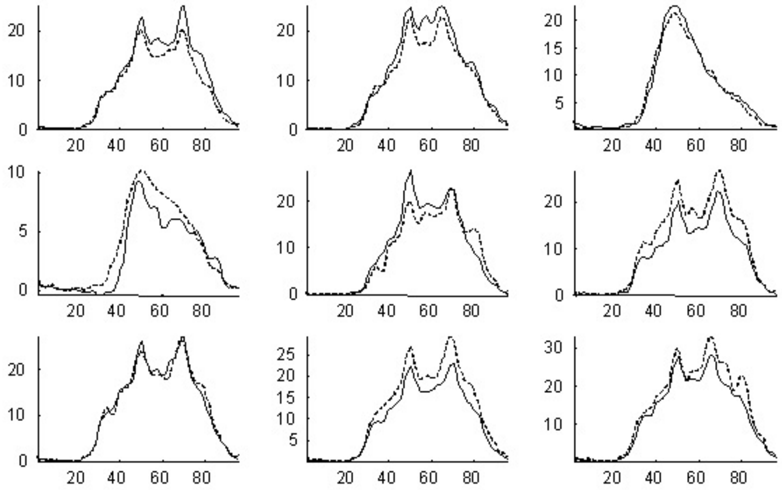
estimated kernel β

Using the estimated model

- Finaly let us discuss the prediction capability of the model.
- Consider the prediction of the function $y_{n+1}(t), t \in [0,1]$.
- According to the model, for each $t \in [0,1]$

$$\widehat{y}_{n+1}(t) = y_{n-6}(t) + \int_0^1 \widehat{\beta}(s,t)[y_n(s) - y_{n-7}(s)]ds.$$

Figure below displays the actual and the forecasted process for nine days. The processes are evaluated every 15 minutes (96 point in x axis).



20
10
0
20 40 60 80
15
5
30
25

- Different types of forecasting errors can be considered.
- The most widely accepted measures of performance of the estimator $\widehat{y}_{n+1}$ is its mean integrated square error given by

$$MISE = E\int_0^1 (y_{n+1}(t) - \widehat{y}_{n+1}(t))^2 dt.$$

  and the mean uniform error

$$\Delta_\infty = E \sup_t |y_{n+1}(t) - \widehat{y}_{n+1}(t)|.$$

- Both these quantities we evaluated by choosing ten days that where not used in the parameter estimation.

- It is clear that the predicted values strongly depends on the number of principal components used in the model.
- The table below shows this dependence via MISE and Δ_∞ errors.

Error\M	1	2	3	4	5	6	7	8
Δ_∞	5.50	5.02	4.99	4.95	5.05	4.97	5.04	5.07
MISE	5.59	4.74	4.73	4.54	4.72	4.76	4.76	4.81

Prediction errors